Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | Remote - US

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python engineering, specializing in **AI agent systems, evaluation harnesses, and tool-calling LLM pipelines**. Built AgentProbe - an open-source agentic AI toolkit with a 3-agent LangGraph runtime (tool calling, structured output, inter-agent state handoffs), a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance), and a 15-endpoint FastAPI backend with CI/CD quality gates. Applied AI at scale across 1.6M+ regulated financial records with output validation and production monitoring. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; designed Tableau, ggplot2, and Quarto dashboards improving faculty research comprehension by 40%.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed, improving downstream model reliability.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Pioneered Entrata's AI engineering foundation -- built and internally shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Built 20+ async Python microservices (FastAPI, PostgreSQL, RabbitMQ) across 120+ Agile sprints for **100K+ monthly active users** -- 40% deployment time reduction; 50% execution time cut via Docker containerization.
- Improved query performance 30% across 200+ SQL queries; achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios.

PROJECTS

AgentProbe: Agentic AI Eval & Testing Toolkit
Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built **evaluation harnesses + quality gates**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) blocks non-compliant agent runs before production; CI/CD integration via GitHub Actions; React dashboard tracks regression drift and retrieval quality across concurrent test suites.
- Architected **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, structured output, inter-agent state handoffs, and multi-provider routing (Anthropic, OpenAI, Groq, Ollama) -- deployed as 15-endpoint FastAPI backend running 100% locally at zero cost.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building a production NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and semantic embeddings -- **HIPAA-compliant RBAC** across structured healthcare records.

Leasel: AI Lease & Affordable-Housing Compliance System

Python · LangGraph · ChromaDB · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_raq

- Automated LIHTC/Section 42 income-eligibility checks previously performed by hand -- LLM extraction feeds a deterministic rules engine that validates lease terms against jurisdiction-specific limits and returns citation-backed results.
- Architected compliance logic outside the model (LLM extracts and explains; rules engine decides) and built a pass@k evaluation harness with a regression gate to measure accuracy.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, Tool Calling, Structured Output, Eval Harnesses, LLM-as-Judge, LLM Evaluation, RAG Pipelines, Agent Observability, Prompt Engineering, ChromaDB, Embeddings, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, Pydantic

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL